SAFETY OF COLLABORATIVE ROBOTS

Manipulator robotics

Abstract

In recent years, industries have sought to automate labor-intensive and repetitive tasks traditionally performed by humans by integrating robots specially designed to execute hazardous industrial tasks with increased precision. Initially, these robots were often deployed in separate workspaces, minimizing interactions with humans but requiring periodic human interaction. Advancements in technology and ongoing research efforts have focused on enabling manipulators to operate alongside humans within the same workspace. To ensure the safety of such interactions, specific safety measures must be followed; otherwise, these manipulators could be hazardous to humans. This report delves into the safety modes typically associated with collaborative robots, explores the essential sensing systems required to meet safety standards, and presents a case study involving Universal Robot's collaborative robot deployed in a real-world application, specifically in the context of welding.

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Contents

List of Tables	2
List of Figures	3
Introduction	4
Safety of Collaborative Robots	5
Current Deployment of Collaborative Robots	9
Case Study	10
Machine Vision	12
Conclusion	13
References	13

List of Tables

Number	Table Description	Page
1	Comparison of Industrial and Collaborative Robots	8

List of Figures

Number	Figure Description	Page
1	Cognitive Human-Robot Interactions	5
2	Illustrating different speeds based on separation of operator from manipulator	7

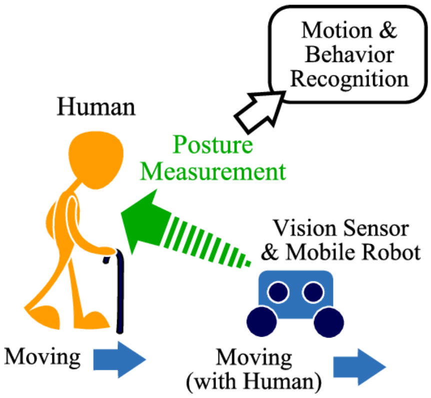
Introduction

Collaborative robots, as defined in **[1]**, are designed to physically interact with humans in a shared workspace. The interaction between humans and these robots, characterized by the robots' relatively low autonomy and complexity, is commonly referred to as Human-Robot Interaction (HRI). Industrial collaborative robots play a crucial role in assisting humans in tasks such as heavy lifting and tracking assembly lines, enhancing productivity by ensuring safety, speed, and accuracy in these operations. The differentiation of cobotics systems involves several key methods:

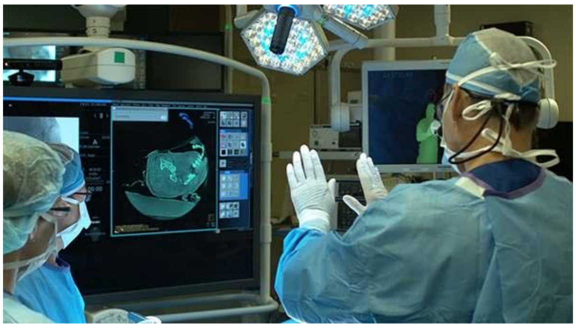
1. *Hardware and Software Design of Collaborative Robots*: The design considerations include framework planning, control structure, and software architecture. These aspects are critical for achieving better cycle control of the manipulator. Additionally, cloud-based and fog-based processing can be employed to enhance the manipulator's capabilities. Special attention may be given to controlling the impedance of the manipulator during the design of its control structure.

2. *Cognitive-Human Interactions*: Effective communication between the operator and the cobot involves recognizing gestures from co-workers. Techniques such as ontologies or randomized forests are utilized to interpret human actions, whether they involve single or multiple actions. External devices, such as enlarged gloves, capture hand gestures, serving as commands to minimize errors. Access control is enhanced through facial, fingerprint recognition, and voice commands, particularly in the local language, ensuring that only authorized personnel can work alongside the cobot. Figure 1 illustrates the different ways of cognitive human-robot interactions.

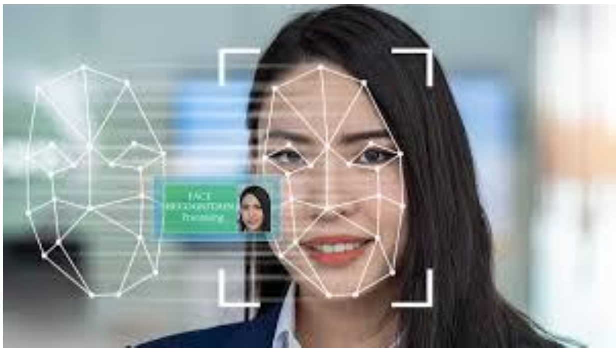
3. *Human-Robot Task Allocations*: Safe and efficient collaboration is achieved when tasks for both the robot and human are well-defined and optimized. Ontology-based knowledge, particularly beneficial for individuals unfamiliar with robotics, simplifies communication and enhances autonomy. High-level task plans facilitate seamless interactions between humans and cobots for both simple and complex tasks. The ability to easily change tasks, coupled with a straightforward task allocation and scheduling process, ensures safer collaboration by clarifying each party's role in the decision-making loop.



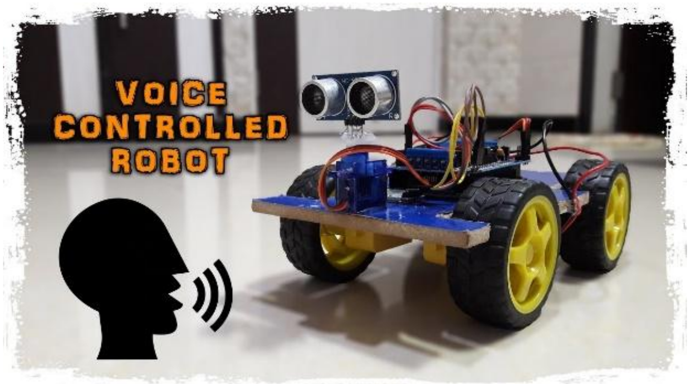
a) Human Action recognition



b) Gestures Recognition



c) Face Recognition



d) Voice Command

Fig 1: Cognitive Human-Robot Interactions

Safety of Collaborative Robots

Robots inherently pose risks to operators, necessitating measures to mitigate these dangers. Traditional industrial robots are typically enclosed by protective fencing to minimize operator exposure to potential hazards, such as unintended movements or end-effector failures. However, instances arise where operators need to enter these cells for tasks like maintenance or inspection, requiring the robot to be disabled, potentially leading to accidents during safeguard deactivation.

The ISO 10218 part 1 and 2 **[2] [3]** provide guidelines for identifying hazards associated with industrial robots, emphasizing safety requirements and protective measures. These may include implementing limiting devices, both mechanical and non-mechanical, to control robot motion and prevent accidents. The risk assessment process outlined in ISO 10218-1:2011 involves considerations such as intended robot operations (teaching, maintenance, setting, cleaning), abrupt start-ups, all-directional access to the robot, potential misuse, control system failures, and application-specific hazards.

ISO 10218-2:2011 specifically addresses collaborative robot applications, offering insights into risk anticipation and reduction. Key steps include iterative processes to assess the effectiveness of risk reduction measures, eliminating overly restrictive measures, and addressing task/hazard pairs with higher initial risks through methods like elimination, substitution, or guarding. Additionally, the standards recognize the need for guarding in non-collaborative tasks using the same equipment. It's crucial to note that these standards, initially designed for industrial robots, are equally applicable to collaborative robots (cobots) used in various industries. Adhering to these standards ensures a comprehensive approach to robot safety, considering different operational scenarios and potential risks. The ISO TS 15066 **[4]** provides technical specifications for modes that a collaborative operation may include. These modes include:

1. *Safety-Rated Monitored Stop*: The cobot in the presence of an obstruction must:
   i. Ensure stop-motion condition
   ii. Ensure Driver power is on
   iii. Ensure that motion resumes after clearance of obstruction
   iv. Ensure the motion resumes without additional action
   v. Protective stop is delivered if stop condition is violated
2. *Hand Guiding*: For an operator to lead the cobot through direct interface:
   i. The robot should stop on the operator's arrival
   ii. Motion should be activated when the operator grasps the enabling device
   iii. Robot motion must respond to operator commands
   iv. Operation resumes on the leave of the operator from the collaborative workspace

3. *Speed and Separation Monitoring*: Figure 2 clearly shows the robot speed that needs to be followed, which is solely based on the distance of the operator from the robot. This distance is generally calculated using scanners, vision systems, or proximity sensors.

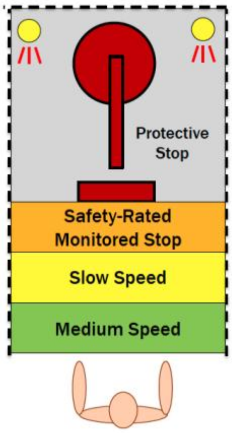


Fig 2: Illustrating different speeds based on separation of operator from manipulator

4. *Power and Force Limiting:* The power and force that can harm a human in case of collision with them has to be limited. This can be achieved by eliminating pinch points, reducing the velocity which in turn reduces the transfer energy, reducing the inertia of the robot, by modifying posture of robot to expose maximum surface area or avoiding sensitive body area such as head and neck.

A review of the ISO 10218 was done by Peter Chemweno, Liliane Pintelon, and Wilm Decre **[5].** They specify the potential sensing systems and types of actuators that can be used for ensuring safety of the operator and basically maintaining the standards set by ISO 10218 and the modes mentioned in ISO TS 15066. The strategies mentioned are:

1. *Passive safeguards:* they include laser scanners, lightweight robotic structure, foam padding. Fluid-based actuators with force and torque limiting functions. Torque and force compliant mechanical actuators.
2. *Active safeguards:* they include intelligent assist device (IAD), impedance control for collision avoidance, agile control-oriented strategies with human cognition, active force controllers using force or torque sensors.
3. *Task Oriented design safeguards:* they include learning from demonstration task design approach. Also include cognitive automation strategies and task-based constraints for motion planning.
4. *Biomechanical models and metrics for quantifying collision risks:* probability and collision impact metrics along with force and torque limiting thresholds.
5. *Design of interactive virtual environments:* laser zoning approach along with human supervisory control strategies and kinetostatic safety fields.

6. *Risk-oriented approaches:* proactive risk profiling and dependability risk assessment models.

The research on kinetostatic safety fields **[6]** serves as an exemplary illustration of an advanced potential sensing system. This system relies on variables such as position, velocity, and body shape and size to effectively assess risk. The closed-form computability offered by this field streamlines real-time risk calculations in the proximity of any arbitrary rigid body. To validate the efficacy of this approach, experiments were conducted using the ABB FRIDA robot.

In a related study **[7]**, human participants were engaged in experiments with collaborative robots, yielding positive feedback. Most participants expressed satisfaction while working with these robots, particularly highlighting positive ratings for interactions such as gesture-based and hand-guided control. This experiment underscored that robots, exemplified by the FourByThree, incorporate safety mechanisms. Operators can be confident that significant precautions are implemented to mitigate risks when sharing a workspace with these manipulators.

Current Deployment of Collaborative Robots:

Over the past couple of years, there has been a significant surge in the utilization of robots across various industries. Primarily, their deployment aims to replace tasks that are repetitive, hazardous, or dangerous, traditionally carried out by humans. Industries contemplating the integration of robots must assess whether collaborative robots (cobots) are necessary for the specific tasks they intend to automate. In comparison to conventional industrial robots, cobots offer several advantages. A study highlighting the distinctions between these two types of manipulators has been conducted by Miriam Matúšová, Marcela Bučányová, and Erika Hrušková **[8]**. Table 1 illustrates some key differences between a Collaborative Robot and a Conventional Industrial Robot. Collaborative Robots Conventional Industry Robots

Collaborative Robots	Conventional Industry Robots
Simple Programming	Programming is time consuming
Lower weigh of robot < 29kg	High Weigh of robot > 50kg
Embedded Security features	Missing of safety sensors
Operation in the limited space	Large Operation space
Lower weigh of machined workpiece	Greater Load capacity
Mobile	Immobile
External Force Sensors	Sensors of External forces are missing
Faster and Simpler Setting	High operation speed in operation
Flexibility of deployment	Universal usage in the limited space

Table 1: Comparison of Industrial and Collaborative Robots [8]

Upon examining the distinctions between the two types of manipulators, it becomes evident that collaborative robots find application in place of conventional industrial robots in scenarios demanding flexibility, straightforward programming, interaction with humans, and where high load capacity is not imperative. Specific industrial applications, such as assembly, surface polishing, palletizing, packing, painting, welding, screwing, engraving, and various other tasks requiring either human-robot interaction or less workload capacity, are suitable domains for the deployment of collaborative robots. A case study mentioning one such scenario is detailed below.

Case Study

b. In this scenario, a collaborative robot is employed for MIG welding to enhance production speed, accuracy, and address manpower constraints. Using a cobot in this application offers two key advantages: optimized division of labor, where robots handle simpler tasks, allowing human workers to focus on more complex welding, and reduced exposure of workers to hazardous welding conditions. Hazards in welding, such as exposure to high heat and chemicals, can have detrimental effects on operators, particularly their eyes. Operators traditionally use protective gear to minimize exposure, but the collaborative robot reduces direct contact with these hazards, thereby lowering the risk of injuries like burns. Choosing a collaborative robot over an industrial robot is justified due to the constant monitoring and frequent changes required in the welding process. Given the risky nature of the application, specific safety standard modes are crucial:

i. ***Safety-Rated Monitored Stop***:

Advantages: This mode enables the operator to inspect the welding without obstructing the cobot. It ensures that the cobot pauses its operation, waiting for the operator to leave the collaborative workspace, preventing over-welding.

Disadvantages: During inspection, since the cobot comes to a stop, there is a pause in the production speed during manufacturing

ii. ***Hand-Guiding***:

Advantages: Hand-guiding mode facilitates operator guidance through challenging locations during task teaching, prioritizing safety.

Disadvantages: The operators sometimes may not be able to teach complicated welding as the cobot can reach a point of singularity during the welding process.

iii. *Speed and Separation Monitoring*:

Advantages: Regular inspection during welding necessitates the implementation of speed and separation monitoring. This mode is essential for the health and safety of the operator. As the primary goal of installing the cobot is to speed up welding tasks while maintaining a collaborative environment, this mode allows the cobot to work at a maximally reduced speed, ensuring safety in the operator's vicinity. The closer the operator gets to the cobot, the slower the speed, meeting the condition of inspection mentioned earlier.

Disadvantages: There are possibilities that there might be a late detection of the presence of an operator by the sensors for a certain distance. This can lead to failure of speed and separation monitoring as its main job is to detect how far the operator is from the cobot and activate the safety-rated monitored stop if they are too close.

Addition of Machine Vision sensors as external Sensors

While the implemented safety measures and modes are effective for collaborative welding, there remains a risk for operators overseeing the task due to potential hazards from heat and chemicals. To mitigate this risk, a viable solution is the installation of a machine vision system alongside the welding tool-piece. This setup allows the operator to inspect the manipulator's welding without being in close proximity to the cobot, acting as a protective measure. Further details on the application of machine vision are provided later in the report, addressing a specific problem and proposing machine vision as a solution.

Machine Vision

Utilizing robots and Artificial Intelligence (AI) for complex problem-solving enhances human-robot interaction, particularly in challenging environments with variable positions of parts, tools, and workers. Employing machine vision techniques, both 2D and 3D **[9]** proves instrumental in addressing such issues. To illustrate, consider the automatic assembly of a car door, involving the placement and screwing of components. The following specific challenges were addressed:

1. *Inner Door Black Module Position Estimation*: The use of a UR10 robot with advanced features, including a Robotiq FT150 force sensor, a tool interchange mechanism, a vacuum, and a screwdriver, along with VRmagic D3 stereo camera and PMD nano time-of-flight (ToF) camera, allowed for accurate estimation of the position of the inner plastic module. The stereo camera identified relevant components based on color, facilitating subsequent tasks.

2. *Door Reconstruction and Position Correction*: Addressing the substantial size of the car door, the system registered various partial views through stereo vision. This process facilitated 3D reconstruction, incorporating depth perception. The door was thus prepared for subsequent assembly tasks, such as screwing.

3. *Screw Detection*: Machine vision, combining 2D techniques with CAD and a 90% threshold, was employed to precisely identify the screws required for the task. This accuracy ensured effective pick-up using the screwdriver tool. Sequence of the Overall Assembling Operation: a) Teaching and Configuration: Nominal points, such as the starting position, locations for retrieving the inner module and screws, and initial detection points, were captured to configure the system. b) Assembling Stage: The robot employed machine vision to scan and collect data for various assembly tasks, including framing and screwing. These data guided the robot in executing precise assembling within the framework. In summary, the integration of a UR10 robot with advanced features and machine vision technologies effectively addressed the challenges associated with the automatic assembly of a car door. This approach ensures efficiency and accuracy in complex and variable environments.

Conclusion

This report provides a brief overview of collaborative robots, their industrial applications, distinctions from conventional robots, the requisite safety standards, operational modes associated with cobots, and a real-world example using a Universal Robot cobot. Additionally, it suggests improvements like using machine vision to enhance safety for the operator and improve collaborative functionality of a cobot.

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